MTH105

Examination - July Semester 2025

Fundamentals of Mathematics

Monday, 10 November 2025 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **SIX (6)** questions and comprises **FOUR (4)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1a

Employ a truth table to determine whether the following argument is valid:

$$p \to (q \vee r) \text{ (premise)}$$
$$r \to (\sim q) \text{ (premise)}$$
$$\therefore r \to p \text{ (conclusion)}$$

(10 marks)

Question 1b

Use the Rules of Inference to show that the following argument form is valid.

$$\forall x\ P(x) \to (Q(x) \vee R(x)) \text{ (premise)}$$
$$\exists x\ P(x) \text{ (premise)}$$
$$\forall x\ \sim Q(x) \text{ (premise)}$$
$$\exists x\ P(x) \wedge R(x) \text{ (conclusion)}$$

(10 marks)

Question 2a

Give a counter-example to show that the following statement is false.

$$\forall x \in \mathbb{N}\ \exists y \in \mathbb{Z}\ \exists z \in \mathbb{Z} \text{ such that } x = y^2 + z^2\,.$$

(5 marks)

Question 2b

Provide the negation of the following statement without using the implication or negation symbols.

$$\exists x \in \mathbb{R}\ \exists y \in \mathbb{Z}\ \forall z \in \mathbb{Q}\,(y < xz \to y + 1 \leq xz)\,.$$

(5 marks)

Question 2c

Let

$$D = \{x \in \mathbb{Z} : -5 \leq x \leq 10\}.$$

Suppose that the variable x has domain D. Determine the truth set of the predicate

$$\forall y \in \mathbb{N}\ \forall z \in \mathbb{Z}\,(x = y^2 z \to y = 1)$$

(5 marks)

Question 3a

Use the Euclidean Algorithm to find $d = \gcd(4181, 962)$ and express d as an integral linear combination of 4181 and 962.

(8 marks)

Question 3b

Let k and p be any positive integers. Show that if $p \neq 1$, then

$$\sum_{i=1}^{k} \frac{1}{p^i} = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}$$

is not an integer.

(7 marks)

Question 4a

Use mathematical induction to show that

$$\sum_{r=1}^{n} \frac{1}{(2r+1)(2r+3)} = \frac{n}{3(2n+3)}$$

for all positive integers n.

(12 marks)

Question 4b

Use the result given in Question 4a to evaluate

$$\sum_{r=3}^{10} \frac{1}{(2r-1)(2r+1)}.$$

(8 marks)

Question 5a

Define the sets

$$A = \{-2, 1, 2, 3, 4, 9, 11\},$$
$$B = \{-3, -1, 0, 1, 2, 3, 4, 7, 8, 10\},$$
$$C = \{0, 2, 3, 5, 6, 7, 8, 11, 12\}.$$

Determine the sets $(A \cup B) \cap C$ and $C \setminus (A \triangle B)$.

(8 marks)

Question 5b

Define the sets

$$D = \{0, 1\},$$
$$E = \{0, 1, 2\}.$$

Let $F = (D \times E) \cap (E \times D)$. List the elements of F and state the number of elements in the power set of F.

(7 marks)

Question 6a

Define a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by

$$f(x) = \frac{x}{|x| + 1}.$$

Determine with justification whether f is injective and/or surjective.

(10 marks)

Question 6b

Define a relation R on the set $\{1, 2, 3, 4, 5, 6\}$ as follows:

$$R = \{(1,1), (2,2), (3,3), (4,4)(5,5), (1,4), (4,1), (2,5), (5,2), (3,6), (6,3)\}$$

Determine whether R is reflexive, whether R is symmetric and whether R is transitive.

(5 marks)

----- END OF PAPER -----